SIMATS ENGINEERING

ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	**DHILIBAN M**
Register Number	**192421166**

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

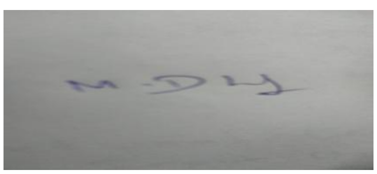
Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

Code of Conduct

I DHILIBAN M/ 192421166 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer

- Infrastructure layer, platform layer, application layer, shared responsibility

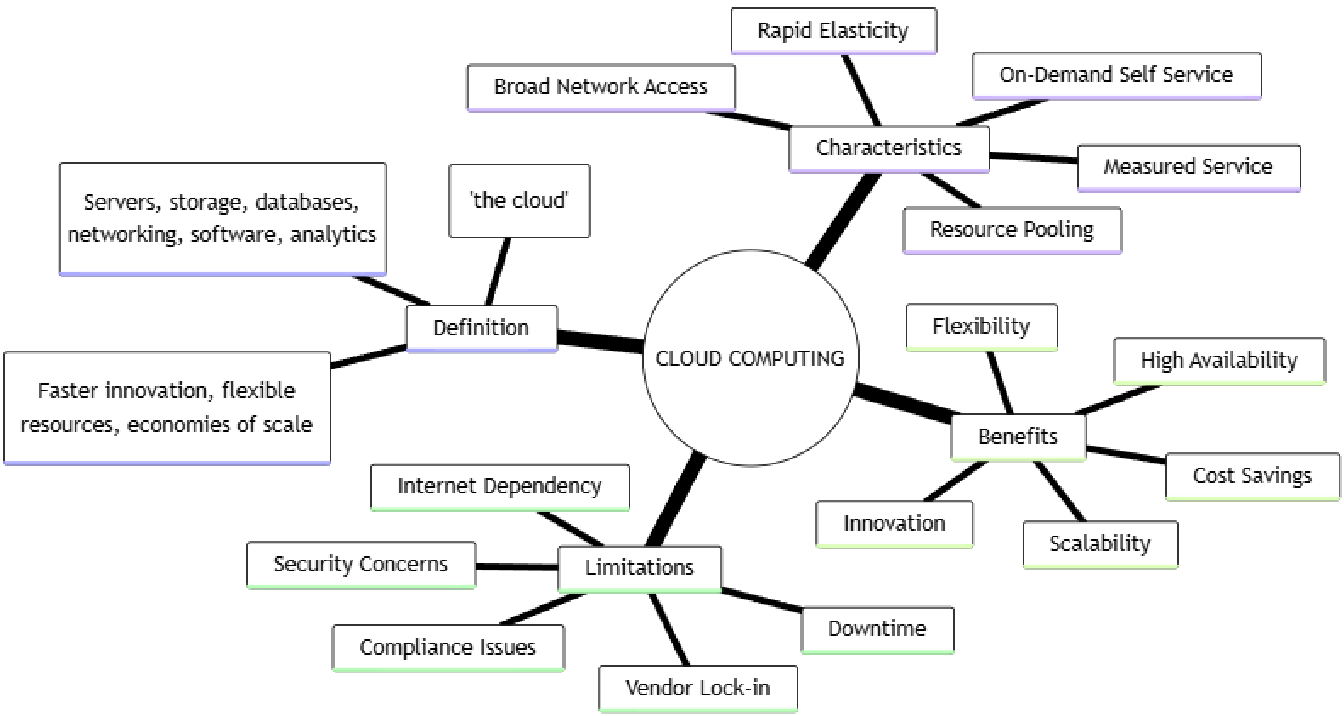
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

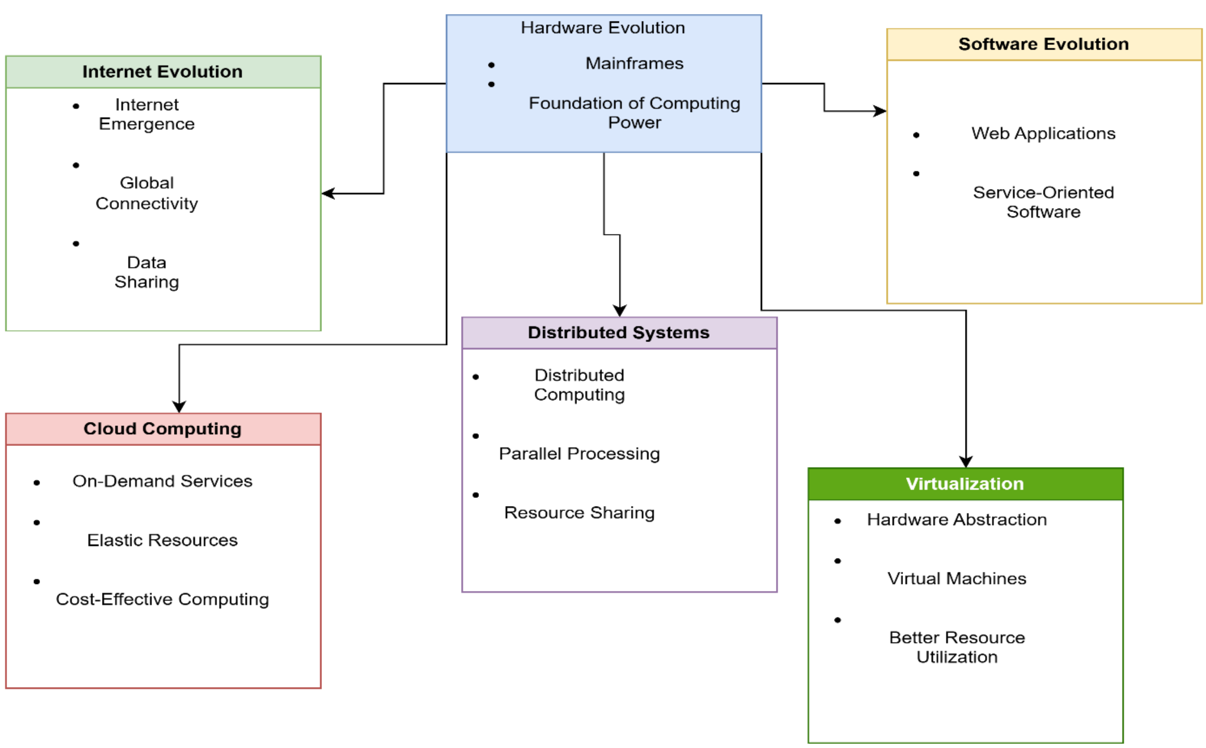
5. Questions

i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.



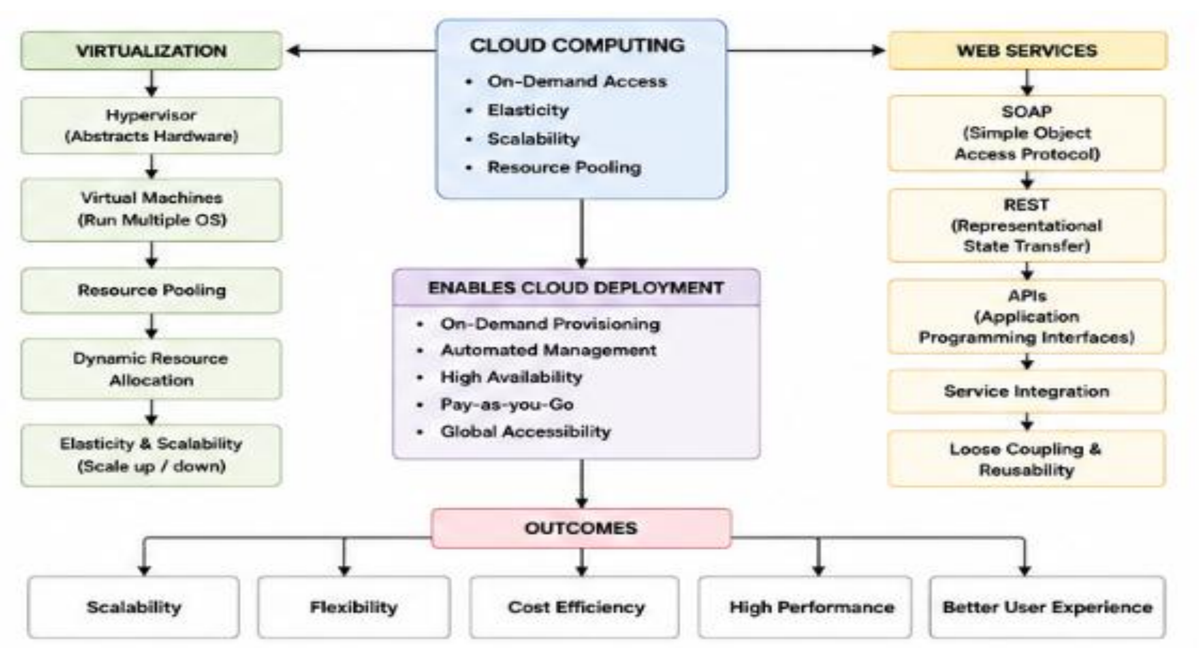
Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.

IT Stack Layers	IaaS Infrastructure as a Service	PaaS Platform as a Service	SaaS Software as a Service	XaaS Anything as a Service
Applications	User Manages	User Manages	Provider Manages	
Data	User Manages	User Manages	Provider Manages	
Runtime	User Manages	Provider Manages	Provider Manages	
Middleware	User Manages	Provider Manages	Provider Manages	
Operating System	User Manages	Provider Manages		

Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.



1. Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?

The concept map effectively represents cloud computing principles by organizing the concepts in a clear hierarchical structure. It begins with cloud computing as the central concept and connects related topics such as characteristics, service models, virtualization, and web services. The use of labeled arrows clearly shows the relationships between concepts. Color coding and logical grouping improve readability and understanding. These design decisions make the concept map simple, accurate, and easy to follow while covering all the important cloud computing concepts.

2. Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Scalability allows the cloud system to handle increasing workloads by adding more resources when needed. Elasticity automatically adjusts resources according to user demand, while resource management ensures efficient allocation of computing resources. These features improve performance, reduce operational costs, and provide a better user experience. Without scalability and elasticity, the system may become slow during high traffic, waste resources during low usage, and experience reduced reliability and availability.

3. Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

REST APIs and SOAP web services are included because they enable communication between different cloud applications and services. REST is lightweight, faster, and suitable for modern web and mobile applications, while SOAP provides secure and reliable communication for enterprise systems. Cloud services such as IaaS, PaaS, and SaaS provide different levels of infrastructure, platform, and software support. Together, these technologies improve interoperability, flexibility, and efficient service delivery.

4. Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

The concept map shows that virtualization supports resource pooling, which improves scalability and elasticity. Web services and APIs enable seamless communication between applications, while service models provide different levels of cloud resources to users. These interactions improve system reliability by ensuring continuous service availability, enhance performance through efficient resource utilization, and provide a better user experience by delivering faster and more flexible cloud services.

5. Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Developing the concept map improved my understanding of cloud computing architecture by showing how different technologies are interconnected. It helped me understand the roles of virtualization, service models, APIs, scalability, and elasticity in building cloud solutions. If I redesigned the concept map, I would include additional cloud security concepts, deployment models such as public, private, and hybrid clouds, and more real-world examples to make the map more informative and practical.